

Autonomous Regulatory Agent (ARA)

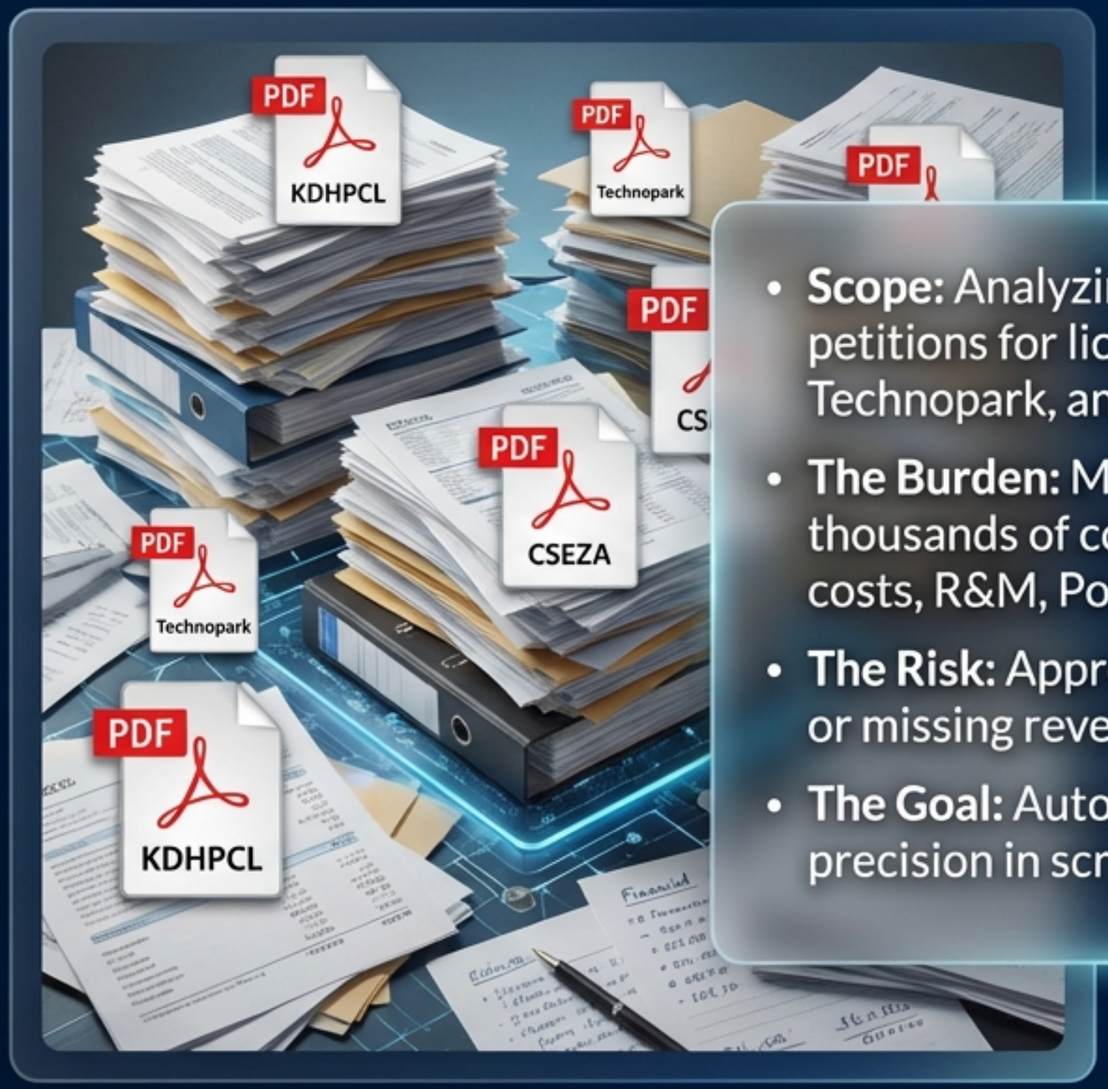
An AI-Based Tool for Smart Truing-Up & Regulatory Compliance

Rajadhani Institute of Engineering and Technology (RIET)

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The Challenge: Regulatory Complexity & Data Overload

Context: Work Order No. KSERC/CSO/03-05 dated 28th January 2026.



- **Scope:** Analyzing massive truing-up petitions for licensees like KDHPCL, Technopark, and Infopark.
- **The Burden:** Manual validation of thousands of cost elements (Employee costs, R&M, Power Purchase).
- **The Risk:** Approving 'imprudent' expenses or missing revenue gaps due to volume.
- **The Goal:** Automated, mathematical precision in scrutiny.



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Meet ARA: Your Virtual Regulatory Auditor



Not a Chatbot.
An Analytical Engine.

1. **Validate:** Verify cost elements against invoices and norms.
2. **Spot:** Identify revenue gaps and deviations (e.g., Distribution loss increases).
3. **Check:** Perform prudence checks against Tariff Regulations.

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The ARA Architecture: A Multi-Agent System

ACTION
(Output)

Generates the 'Analysis and Decision' draft.



AGENTS
(Analysis)

Specialized modules that 'think' like different staff members.

RETRIEVAL
(Knowledge Base)

Ingests The Electricity Act, 2003 & Tariff Regulations.

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Layer 1: The Data Specialists

The Clerk
(Data Ingestor)



Pulls data directly from petitions. Maps unstructured text to Regulatory Data Schema.

The Trend Analyzer
(Forensic Auditor)



Spots deviations. Flags expenses exceeding inflation indices or historical averages.

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Layer 1: The Compliance Specialists

The Legal Brain
(Compliance)



Conducts prudence checks against Tariff Regulations 2021. Distinguishes Controllable vs. Uncontrollable costs.

The Secretary
(Reporter)



Drafts the ‘Analysis and Decision’ chapter. Cites specific table numbers and disallowances.

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R - Retrieval: A Robust RAG System

Vector DB (Unstructured)



Electricity Act 2003,
Tariff Regulations 2021.

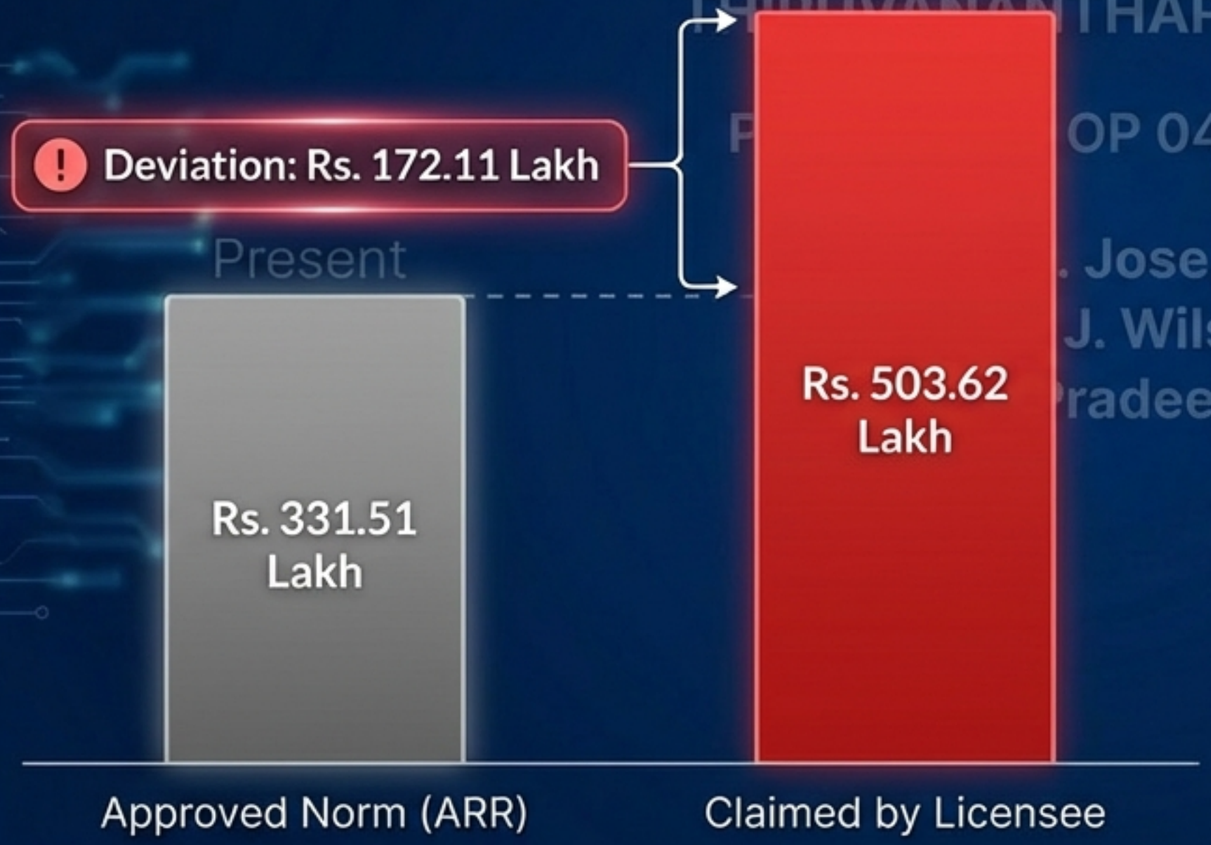
SQL DB (Structured)



Historical Truing Up Orders,
Inflation Indices (CPI/WPI).

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The Forensic Auditor in Action: Technopark R&M Scrutiny.



High Alert

ARA Flag: Massive variance identified. Contracts like 'Fathima Engineering' flagged for manual review.

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The Legal Brain in Action: Employee Cost Validation.

Compliance Checklist

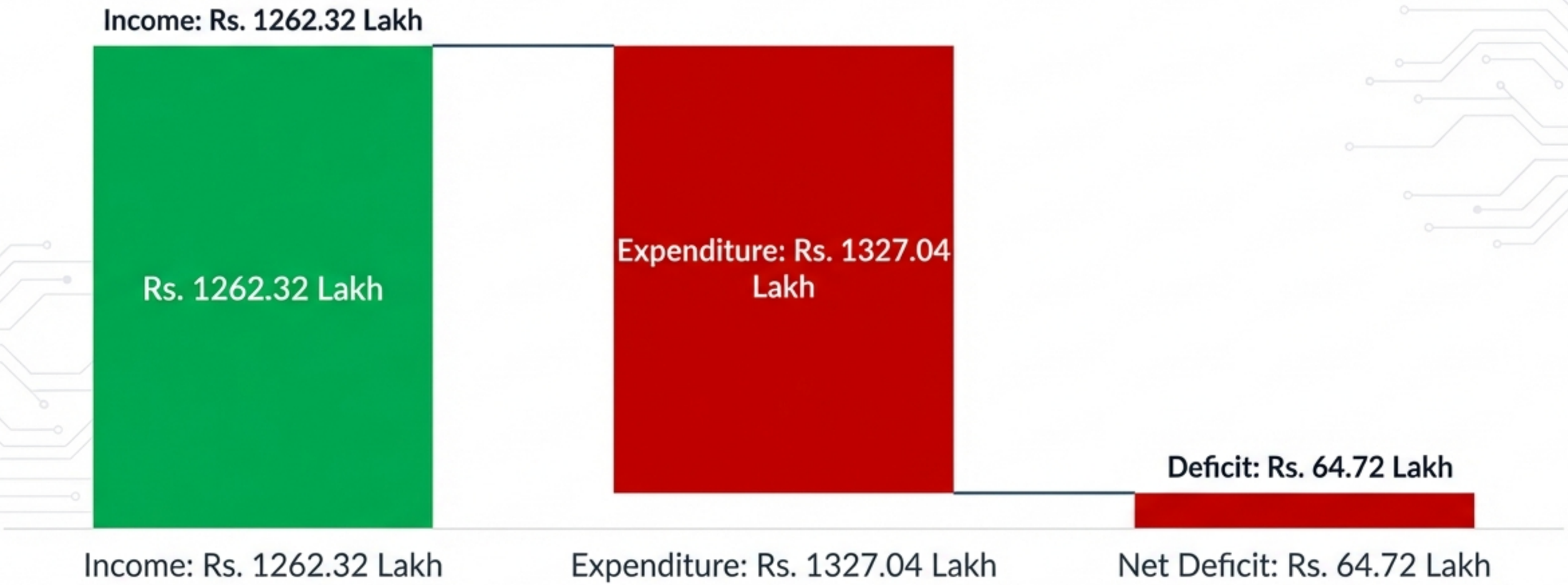
Item	Inter	Lato
1.	Licensee: **CSEZA**	**(2023-24)**
2.	Regulation Norm:	**Rs. 166.65 Lakh**
3.	Claimed Amount:	**Rs. [illegible] Lakh**



Note: Claim is valid, but efficiency gains must be calculated per Regulation 18.

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The Brain in Action: Truing Up Math (Infopark)



Cumulative Gap Updated: Rs. 915.59 (Opening) + Rs. 64.72 (Current) = **Rs. 980.31 Lakh**

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Layer 3 - Action: Automated Order Drafting

Input



Automated Drafting

Output

Draft Order

The Commission notes that the R&M expense claimed (Rs. 503.62 lakh) is higher than the approved ARR norms (Rs. 331.51 lakh). The licensee is directed to limit expenses to reasonable levels.

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Future Capability: Resource Adequacy Planning

Beyond Truing Up

LOLP Calculation (Loss of Load Probability)

Capacity Credits for Solar/Wind using Draft Regulation Framework



$$RAR = \sum (\text{Solar Capacity} * \text{Capacity Credit}) + \sum (\text{Wind Capacity} * \text{Capacity Credit})$$

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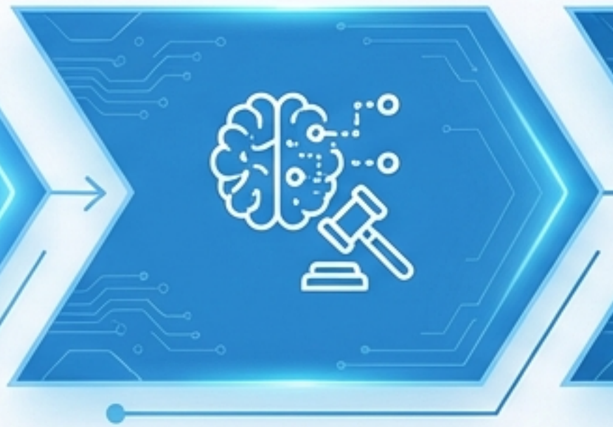
Implementation Roadmap

Phase 1:
Weeks 1-2



Data Ingestion (PDFs to SQL).

Phase 2:
Weeks 3-5



Agent Training (Fine-tuning on KSERC Orders).

Phase 3:
Weeks 6-7



UAT (User Acceptance Testing).

Phase 4:
Week 8



Final Deployment.

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The Value Proposition



Speed

Reduces scrutiny time from weeks to hours.



Accuracy

Eliminates human error in Revenue Gap calculations.



Compliance

Prevents ‘Double Payment’ on assets created from regulatory surplus.

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Empowering KSERC with Intelligent Oversight

Moving from Data Entry to Data Decision.



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